

Nonintegral Laplacian spectra of ideal intersection graphs

Working manuscript

October 4, 2026

Abstract

We study Laplacian integrality for the graph of nonzero proper ideals of $\mathbb{Z}_n$, where distinct ideals are adjacent when their intersection is nonzero. For squarefree composite n , the graph is Laplacian integral exactly when n has two prime factors. For three distinct primes we prove nonintegrality for exponent vectors $(1, 1, k)$, $k \geq 1$, and (a, a, b) whenever $a \geq 2$ and $b > 2(a - 4)(a - 2)/(3a + 4)$. The repeated-exponent problem admits an exact finite divisor reduction for each fixed a , and also for each fixed smaller-factor index; we settle every $b \geq 1$ when $2 \leq a \leq 9$. The proofs use small invariant subspaces, a cubic obstruction on a boundary family, and arithmetic constraints on a quadratic discriminant.

1 Introduction

The ideal intersection graph associates a graph to the intersection structure of the nonzero proper ideals of a ring [1]. For $\mathbb{Z}_n$, its isomorphism type depends on the exponents in the prime factorization of n . We ask which of these graphs have integer Laplacian eigenvalues.

We prove a complete classification for squarefree composite integers (Theorem 3.1), an explicit quadratic obstruction for $(1, 1, k)$ (Theorem 4.1), and a linear cutoff for repeated exponents (Theorem 6.7). The cutoff combines an arithmetic restriction on the antisymmetric block with the symmetric cubic obstruction for $b = (a - 1)(2a - 1)$ proved in Theorem 5.1. An exact divisor criterion isolates every possible square discriminant for fixed a . A second divisor criterion fixes the smaller-factor index instead, giving a finite reduction over all a (Proposition 6.6). Together these give nonintegrality for all b when $2 \leq a \leq 9$ (Theorem 7.1 and Corollaries 7.2, 7.3).

The graph and lifting arguments are given explicitly, so the proofs do not require a complete spectral decomposition. Section 8 describes the exact computations supporting the displayed formulas and finite certificates. The classification for arbitrary exponent vectors remains open.

2 The graph and its invariant subspaces

Write $n = \prod_{i=1}^t p_i^{k_i}$, where the primes are distinct and $k_i \geq 1$. Denote by G_n the graph whose vertices are the nonzero proper ideals of $\mathbb{Z}_n$, with distinct ideals adjacent if their intersection is nonzero. We use the combinatorial Laplacian $L = D - A$; the graph is *Laplacian integral* when every eigenvalue of L is an integer, including when the graph is disconnected.

The ideal $(\prod_i p_i^{\alpha_i})$ is encoded by $0 \leq \alpha_i \leq k_i$. We exclude $(0, \dots, 0)$, which encodes the whole ring, and $(k_1, \dots, k_t)$, which encodes the zero ideal. Ideal intersection is generated by the least common multiple. Therefore two distinct vertices α, β are adjacent exactly when

$$\max(\alpha_i, \beta_i) < k_i \quad \text{for some } i.$$

Equivalently, putting $S(\alpha) = \{i : \alpha_i < k_i\}$, adjacency is $S(\alpha) \cap S(\beta) \neq \emptyset$.

For nonempty $S \subseteq [t]$, the support class $\mathcal{C}_S$ is a clique of size $\prod_{i \in S} k_i$, except that the full support has size $u = \prod_i k_i - 1$. Edges between distinct support classes are complete or absent according as their supports intersect or are disjoint. The full support consists of universal vertices. Remove them and call the resulting graph H ; thus $G_n = K_u \vee H$, with the evident interpretation for $u = 0$.

Lemma 2.1 (Lifting across the universal vertices). *If $L(H)v = \lambda v$ and $\lambda \neq 0$, extension of v by zero on the u universal vertices is an eigenvector of $L(G_n)$ with eigenvalue $\lambda + u$.*

Proof. Since $L(H)$ is symmetric with zero row sums, $\lambda \neq 0$ implies $\sum v = 0$. At a vertex of H , its degree increases by u and all new neighbor values are zero. At a universal vertex the Laplacian action is $-\sum v = 0$. These are exactly the claimed eigenvector equations. $\square$

For $(k_1, k_2, k_3) = (a, a, b)$, H has six support classes of sizes a, a, b, a^2, ab, ab on $1, 2, 3, 12, 13, 23$, respectively. The partition into

$$X = \mathcal{C}_1 \cup \mathcal{C}_2, \quad Y = \mathcal{C}_3, \quad Z = \mathcal{C}_{12}, \quad W = \mathcal{C}_{13} \cup \mathcal{C}_{23}$$

is equitable, and on functions constant on these cells the Laplacian acts by

$$B = \begin{pmatrix} a^2 + ab & 0 & -a^2 & -ab \\ 0 & 2ab & 0 & -2ab \\ -2a & 0 & 2a + 2ab & -2ab \\ -a & -b & -a^2 & a^2 + a + b \end{pmatrix}. \quad (1)$$

For example, an X vertex has a^2 neighbors in Z and ab in W , while a W vertex has a, b, a^2 neighbors in X, Y, Z . Internal neighbors cancel for cell-constant functions. The cell sizes are $2a, b, a^2, 2ab$, and multiplying B on the left by their diagonal matrix gives a symmetric matrix. Thus B is similar to a real symmetric matrix.

The functions taking values $s, -s$ on $\mathcal{C}_1, \mathcal{C}_2$, $t, -t$ on $\mathcal{C}_{13}, \mathcal{C}_{23}$, and zero elsewhere form a second invariant subspace, with matrix

$$B_- = \begin{pmatrix} a^2 + ab & -ab \\ -a & a^2 + a + b + 2ab \end{pmatrix}. \quad (2)$$

These functions have sum zero. Their eigenvalues therefore also lift by $u = a^2b - 1$, directly by the proof of Lemma 2.1.

Expanding (1) gives $\det(xI - B) = xf_{a,b}(x)$, where

$$\begin{aligned} f_{a,b}(x) &= x^3 - E_1x^2 + E_2x - E_3, \\ E_1 &= 2a^2 + 5ab + 3a + b, \\ E_2 &= a^4 + 7a^3b + 3a^3 + 8a^2b^2 + 11a^2b + 2a^2 + 3ab^2 + 2ab, \\ E_3 &= 2a^2b(a + b + 1)(a^2 + 2ab + 2a + b). \end{aligned} \quad (3)$$

Its roots are real and nonzero: reality follows from the similarity above, and $E_3 > 0$. Every root λ gives an eigenvalue $\lambda + u$ of G_n . A noninteger root of $f_{a,b}$ therefore proves nonintegrality.

3 Squarefree integers

Theorem 3.1. *For squarefree composite n , the graph G_n is Laplacian integral if and only if n has exactly two prime factors.*

Proof. Let t be the number of prime factors. For $t = 2$ the graph consists of the two ideals (p) and (q) , with no edge, so $L = 0$. Assume $t \geq 3$. The vertices are the nonempty proper subsets of $[t]$, adjacent when they intersect. On the tensor space encoding all subsets put

$$E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad c = 2^t - 1.$$

The principal submatrix of $\Phi = cI - E^{\otimes t} + F^{\otimes t} - J^{\otimes t}$ on the graph vertices is $L(G_n)$: the diagonal entry at S is $2^t - 2 - 2^{t-|S|}$, and the off-diagonal entry is -1 for intersecting subsets and zero otherwise.

With e_0, e_1 denoting the absence and presence basis, set $w = e_0 \otimes e_1 - e_1 \otimes e_0$. For every 2×2 matrix T , one has $T^{\otimes 2}w = (\det T)w$. A paired factor w thus contributes $2, -1, 0$ under E, F, J , respectively. It vanishes at the empty and full subsets. Each subspace used below is invariant for Φ and is supported on graph vertices, so its restriction is also a restriction of L .

If $t = 2r + 1$, $r \geq 1$, use $w^{\otimes r} \otimes \text{span}\{e_0, e_1\}$. Putting $a = 2^r$ and $s = (-1)^r$, the restricted matrix is

$$\begin{pmatrix} c - 2a + s & s \\ s & c - a \end{pmatrix},$$

with discriminant $(a - s)^2 + 4$. The integer $h = a - s$ is at least 3, so $h^2 < h^2 + 4 < (h + 1)^2$. Both eigenvalues are irrational.

If $t = 2r$, $r \geq 2$, use $r - 1$ paired factors and the symmetric subspace on the last two coordinates, with basis $e_0e_0, (e_0e_1 + e_1e_0)/\sqrt{2}, e_1e_1$. Put $a = 2^{r-1}$ and $s = (-1)^{r-1}$. The restriction is $cI - K$, where

$$K = \begin{pmatrix} 4a - s & -s\sqrt{2} & -s \\ -s\sqrt{2} & 2a - s & 0 \\ -s & 0 & a \end{pmatrix}.$$

The leading principal minors of $K - (a - 1)I$ are

$$3a + 1 - s, \quad (3a + 1 - s)(a + 1 - s) - 2, \quad (3a - s)(a + 1 - s) - 2.$$

They are positive for $a \geq 2$ and $s = \pm 1$: the latter two are at least 10 and 8, respectively. Sylvester's criterion gives $\lambda_{\min}(K) > a - 1$. The principal submatrix of $K - aI$ on coordinates 1, 3 is $\begin{pmatrix} 3a-s & -s \\ -s & 0 \end{pmatrix}$, with determinant -1 . A vector supported on those coordinates has a negative Rayleigh quotient, giving $\lambda_{\min}(K) < a$. Hence $c - \lambda_{\min}(K)$ is an eigenvalue of L strictly between the consecutive integers $c - a$ and $c - a + 1$. $\square$

4 The family pqr^k

Theorem 4.1. *For distinct primes p, q, r and every $k \geq 1$, the graph G_{pqr^k} has the irrational eigenvalues*

$$\frac{6k + 1 \pm \sqrt{(2k + 2)^2 - 3}}{2}.$$

In particular, every permutation of the exponent vector $(1, 1, k)$ gives a nonintegral Laplacian spectrum.

Proof. The support classes 1, 2 have size 1, and 13, 23 have size k . Consider functions with values $s, -s$ on the first two classes, $t, -t$ on the latter two, and zero elsewhere. A vertex of support 1 has $2k$ neighbors and Laplacian value $2ks - kt$. A vertex of support 13 has degree $4k$; after subtracting the same-class and opposite-class values its Laplacian value is $-s + (4k + 1)t$. The swapped classes give their negatives, and all other classes give zero by cancellation. The restricted matrix of the full graph is therefore

$$B_k = \begin{pmatrix} 2k & -k \\ -1 & 4k + 1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - (6k + 1)x + 8k^2 + k$, with discriminant $(2k + 2)^2 - 3$. Since

$$(2k + 1)^2 < (2k + 2)^2 - 3 < (2k + 2)^2$$

for $k \geq 1$, the two roots are irrational. Relabeling prime coordinates preserves the graph. $\square$

5 A boundary family with integral antisymmetric block

The following argument was supplied by Reza Nikandish in the public collaboration discussion. We include it because it is the boundary case needed for the linear cutoff in Section 6.

Theorem 5.1. *For every $a \geq 2$, put $C = (a - 1)(2a - 1)$. For distinct primes p, q, r , the graph $G_{p^a q^a r^a C}$ is not Laplacian integral, although the antisymmetric block (2) has integer eigenvalues.*

Proof. Specialize $f_{a,b}$ from (3) to $b = C$. For $a = 2$ it is $x^3 - 47x^2 + 702x - 3312$. Its values at 0, 1, 2, 3, 4 modulo 5 are 3, 4, 2, 3, 3. Thus it has no root in $\mathbb{F}_5$ and is irreducible over $\mathbb{Q}$; its real roots are noninteger. Lemma 2.1 gives noninteger graph eigenvalues.

For $a \geq 3$ put $m = 2a^3 - 2a^2 + 2a - 1$. Expansion gives

$$\begin{aligned} f_{a,C}(m) &= 2(5a^4 - 10a^3 + 7a^2 - 1), \\ f_{a,C}(m - 1) &= -2(a - 1)(a + 2)(2a^4 - 7a^3 + 7a^2 - a - 3). \end{aligned}$$

The first value is positive because its parenthesis is $5a^3(a - 2) + 7a^2 - 1$. Writing $z = a - 3 \geq 0$, the last factor in the second value is $2z^4 + 17z^3 + 52z^2 + 68z + 30 > 0$. Consequently $f_{a,C}(m - 1) < 0 < f_{a,C}(m)$, so a real root lies strictly between $m - 1$ and m . Its integer shift by $u = a^2 C - 1$ is noninteger.

The characteristic discriminant of B_- is $D = (a + 1)^2 b^2 + 2a(3a + 1)b + a^2$. At $b = C$ it equals $(2a^3 - a^2 + a - 1)^2$, and the two eigenvalues are $4a^3 - 3a^2 + a$ and $2a^3 - 2a^2 + 1$. Thus the obstruction in this family comes from the symmetric cubic. $\square$

6 Finite reduction and a linear cutoff

Fix $a, b \geq 1$, and put

$$h = a + 1, \quad C = (a - 1)(2a - 1), \quad M = a^3(2a + 1), \quad A = h^2 b + a(3a + 1).$$

The characteristic discriminant of (2) is

$$D = h^2 b^2 + 2a(3a + 1)b + a^2. \tag{4}$$

If D is not a square, this block has irrational eigenvalues, which remain irrational after the integer shift $a^2 b - 1$.

Proposition 6.1 (Exact divisor reduction). *For fixed $a \geq 1$, the positive integers b for which D is a square are exactly those obtained from positive factor pairs $d \leq e$ of M such that*

$$b = \frac{d + e - a(3a + 1)}{h^2} \in \mathbb{Z}_{\geq 1}, \quad s = \frac{e - d}{h} \in \mathbb{Z}_{\geq 0}. \quad (5)$$

In that case $D = s^2$.

Proof. The polynomial identity

$$A^2 - h^2 D = 4M \quad (6)$$

holds for arbitrary a, b . If $D = s^2$ with $s \geq 0$, then $A - hs$ and $A + hs$ are positive: their product is positive and their sum is $2A > 0$. They have the same parity; since their product is divisible by 4, both are even. Write $A - hs = 2d$ and $A + hs = 2e$. Then $de = M$, $d + e = A$ and $e - d = hs$, giving (5). Conversely those conditions yield $A = d + e$ and $hs = e - d$. Using $(d + e)^2 - (e - d)^2 = 4de = 4M$ in (6) gives $D = s^2$. $\square$

The divisor set is finite for every fixed a ; this criterion has no truncation in b . It gives first a quadratic bound.

Corollary 6.2. *If D is a square then $b \leq C$. Consequently every $b > C$ gives a nonintegral Laplacian spectrum. For $a = 1$, every $b \geq 1$ is covered.*

Proof. Since $(d - 1)(e - 1) \geq 0$, $A = d + e \leq M + 1$. The identity

$$M + 1 - a(3a + 1) = h^2 C$$

then gives $b \leq C$. The nonsquare case is covered by the antisymmetric block. At $a = 1$, $C = 0$. $\square$

Lemma 6.3. *Every admissible factor pair in Proposition 6.1 satisfies $d \equiv e \equiv 1 \pmod{h}$.*

Proof. From $e - d = hs$ and $d + e = A$, reduction modulo h gives $e \equiv d$ and $2d \equiv 2$, since $a(3a + 1) \equiv 2 \pmod{h}$. Also

$$(d - 1)(e - 1) = M - A + 1 = h^2(C - b). \quad (7)$$

If h is odd, the congruence $2(d - 1) \equiv 0$ gives the assertion. If h is even, the alternative to $d \equiv 1$ is $d - 1 \equiv h/2 \pmod{h}$. Then $d - 1$ and $e - 1$ would both equal $h/2$ times an odd integer. Their product divided by h^2 would be an odd integer divided by 4, contradicting (7). This excludes the alternative for either parity of a . $\square$

Theorem 6.4 (Unified linear cutoff). *Let $a \geq 2$ and*

$$R(a) = \frac{2(a - 1)(a - 2)}{a + 2} = 2a - 10 + \frac{24}{a + 2}.$$

For every integer $b > R(a)$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved repeated-exponent pair therefore has square D and $1 \leq b \leq \lfloor R(a) \rfloor$.

Proof. If D is nonsquare the antisymmetric block suffices. Otherwise use Proposition 6.1. When $d = 1$, $e = M$ and $b = C$; Theorem 5.1 settles this case. If $d > 1$, Lemma 6.3 gives $d \geq T = a + 2$. Since $e \geq d \geq T$,

$$\frac{M}{T} + T - (d + e) = (d - T) \left(\frac{e}{T} - 1 \right) \geq 0.$$

Hence

$$b \leq \frac{M/T + T - a(3a + 1)}{h^2} = \frac{2(a - 1)(a - 2)}{a + 2} = R(a).$$

Thus $b > R(a)$ leaves either a nonsquare discriminant or the already settled boundary case. $\square$

At $(a, b) = (4, 2)$ the square-discriminant pair has $d = 6 = a + 2$, so equality in the factor estimate is attained. The graph is still nonintegral, as shown in the next section.

6.1 Settling the first nonboundary factor

The first possible value after $d = 1$ is $d = a + 2$. Since d divides M , the identity

$$M = (a + 2)(2a^3 - 3a^2 + 6a - 12) + 24$$

forces $a + 2 \mid 24$. With $a \geq 2$ this leaves $a = 2, 4, 6, 10, 22$. Substitution into (5) gives $b = 0, 2, 5, 12, 35$, respectively. Excluding $b = 0$, the complete positive- b list and its cubic certificates are:

(a, b)	$f_{a,b}(x)$	Root-free prime
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(6, 5)$	$x^3 - 245x^2 + 19266x - 488160$	7
$(10, 12)$	$x^3 - 842x^2 + 230160x - 20534400$	13
$(22, 35)$	$x^3 - 4919x^2 + 7887858x - 4132479120$	17

Each polynomial has no root modulo its indicated prime. Its monic cubic is therefore irreducible over $\mathbb{Q}$, and its real nonzero roots lift to noninteger graph eigenvalues. Thus every possible $d = a + 2$ pair is settled, over all a , by this finite divisor argument.

Theorem 6.5 (Second linear cutoff). *For $a \geq 2$ put*

$$S(a) = \frac{(a - 3)(2a - 3)}{2a + 3} = a - 6 + \frac{27}{2a + 3}.$$

For every positive integer $b > S(a)$ and distinct primes p, q, r , $G_{p^a q^a r^b}$ is not Laplacian integral. Every remaining square-discriminant case has $d \geq 2a + 3$ and $b \leq \lfloor S(a) \rfloor$.

Proof. Nonsquare D is covered by the antisymmetric block. If D is square, the cases $d = 1$ and $d = a + 2$ are already settled. Every other admissible factor satisfies $d \geq T = 2a + 3$ by Lemma 6.3. Since $e \geq d \geq T$, the same factor inequality gives

$$b \leq \frac{M/T + T - a(3a + 1)}{(a + 1)^2} = \frac{(a - 3)(2a - 3)}{2a + 3} = S(a).$$

This proves the claim for $b > S(a)$. □

The new bound is strictly smaller than $R(a)$ for all $a \geq 2$, since

$$R(a) - S(a) = \frac{2a^3 - a^2 - a - 6}{(a + 2)(2a + 3)} > 0.$$

Writing $z = a - 2 \geq 0$, its numerator is $2z^3 + 11z^2 + 19z + 4$.

6.2 A finite reduction for each factor index

For a nonboundary square-discriminant pair, write $d = 1 + hj$, $j \geq 1$. The following criterion fixes j instead of a .

Proposition 6.6. *Fix $j \geq 1$ and set $K_j = (j+1)^3(j+2)$. Every admissible square-discriminant pair of index j is obtained from a positive divisor d of K_j with $a = (d-1)/j - 1 \in \mathbb{Z}_{\geq 2}$. Set $e = M/d$, $v = (e-1)/(a+1)$, and retain exactly those candidates with $d \leq e$ and $b = C - jv \geq 1$. Then e, v, b are integers and $D = (v-j)^2$. Conversely every retained candidate is admissible.*

Proof. Modulo $d = j(a+1) + 1$ we have $ja \equiv -(j+1)$, whence

$$j^4 M = (ja)^3 j(2a+1) \equiv (j+1)^3(j+2) = K_j \pmod{d}.$$

Since $\gcd(d, j) = 1$, we have $d \mid M$ if and only if $d \mid K_j$. Thus e is integral. Since $M \equiv d \equiv 1 \pmod{h}$, also $e \equiv 1 \pmod{h}$, so v is integral. The identity

$$M = 1 + h(3a-2) + h^2 C = (1+hj)(1+hv)$$

gives $j + v + hjv = 3a - 2 + hC$. Defining $b = C - jv$, we obtain $d + e = h^2 b + a(3a+1)$ and $e - d = h(v-j)$. The ordering condition gives $v - j \geq 0$; Proposition 6.1 now proves admissibility and $D = (v-j)^2$. The reverse direction follows by applying these identities to any admissible index- j pair. $\square$

This is a complete finite problem for each fixed j , without a truncation in a . It leaves the family over arbitrary j open. For $j = 2$, $K_2 = 108$ and $d = 2a + 3$ is an odd divisor of 108 with $d \geq 7$. The only candidates are $d = 9, 27$, giving $a = 3, 12$. The first has $e = 21, v = 5, b = 0$ and is excluded. The second has $e = 1600, v = 123, b = 7$ and $\sqrt{D} = 121$. Thus $(12, 7)$ is the only positive- b case with $j = 2$.

Its symmetric cubic is $f_{12,7}(x) = x^3 - 751x^2 + 180348x - 13829760$. Its values at all residues modulo 13 are 4, 7, 9, 3, 8, 4, 10, 6, 11, 5, 7, 10, 7. As a monic cubic with no root modulo 13, it is irreducible over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues, settling the entire index-2 case.

Theorem 6.7 (Third linear cutoff). *For every $a \geq 2$, put $T(a) = 2(a-4)(a-2)/(3a+4)$. If $b \geq 1$ and $b > T(a)$, then $G_{p^a q^{a+b}}$ is not Laplacian integral. Every unresolved square-discriminant pair has $j \geq 3$, $d \geq 3a+4$, and $b \leq \lfloor T(a) \rfloor$.*

Proof. Nonsquare D gives irrational antisymmetric eigenvalues. For square D , the cases $j = 0, 1, 2$ are all settled above. Every other pair has $d \geq Q = 3a+4$, and $e \geq d \geq Q$ gives

$$b \leq \frac{M/Q + Q - a(3a+1)}{(a+1)^2} = \frac{2(a-4)(a-2)}{3a+4} = T(a).$$

This proves the claimed implication. $\square$

For $a \geq 3$ the new bound is strictly below $S(a)$, since

$$S(a) - T(a) = \frac{2a^3 - a^2 - 5a - 12}{(2a+3)(3a+4)} > 0.$$

With $z = a - 3 \geq 0$, the numerator is $2z^3 + 17z^2 + 43z + 18$. The exact expression $T(a) = 2a/3 - 44/9 + 320/(9(3a+4))$ shows its leading term.

7 Complete fixed-exponent families and exact diagnostics

Theorem 7.1. *For every $a \in \{2, 3, 4\}$, every $b \geq 1$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral.*

Proof. The cutoff values are $R(2) = 0$, $R(3) = 4/5$, and $R(4) = 2$. Theorem 6.4 settles all b for $a = 2, 3$, and all $b > 2$ for $a = 4$. At $(a, b) = (4, 1)$, $D = 145$ is nonsquare. At $(4, 2)$ the symmetric cubic is

$$f_{4,2}(x) = x^3 - 86x^2 + 2304x - 18816.$$

Its values at $0, 1, 2, 3, 4$ modulo 5 are $4, 3, 1, 4, 3$, so it has no root modulo 5. A cubic over a field without a root is irreducible. Since this cubic is monic, reduction modulo 5 proves irreducibility over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues. $\square$

Corollary 7.2. *The same all- b conclusion holds for $a = 5, 6$. Thus it holds for every $a \in \{2, 3, 4, 5, 6\}$.*

Proof. The second cutoff gives $S(5) = 14/13$ and $S(6) = 9/5$, leaving only $b = 1$ in each case. The discriminants are 221 and 313, respectively. Since $14^2 < 221 < 15^2$ and $17^2 < 313 < 18^2$, both give irrational antisymmetric eigenvalues. Every larger b is covered by Theorem 6.5. $\square$

Corollary 7.3. *Every positive b also gives a nonintegral spectrum for $a = 7, 8, 9$. Thus every repeated exponent $a \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ is settled.*

Proof. The third cutoff gives $T(7) = 6/5$, $T(8) = 12/7$, $T(9) = 70/31$. Only four pairs remain under these bounds:

$$(a, b) = (7, 1), (8, 1), (9, 1), (9, 2).$$

Their discriminants are 421, 545, 685, 1489, respectively, between the consecutive squares with lower square roots 20, 23, 26, 38. Each therefore gives irrational antisymmetric eigenvalues, and Theorem 6.7 covers every larger b . $\square$

For comparison, the complete divisor certificates before applying the linear cutoff are given in Table 1. The constants factor as $M = 40 = 2^3 \cdot 5$, $189 = 3^3 \cdot 7$, and $576 = 2^6 \cdot 3^2$. Testing every divisor $d \leq \sqrt{M}$ in (5) gives exactly the displayed admissible pairs. The corresponding cubics in Table 2 have no root modulo the listed primes. These finite certificates supply a second proof using the complete exceptional list. Three pairs are in the boundary family; $(4, 2)$ is the additional symmetric-block obstruction.

a	M	(d, e)	b	$\sqrt{D}$
2	40	(1, 40)	3	13
3	189	(1, 189)	10	47
4	576	(6, 96)	2	18
4	576	(1, 576)	21	115

Table 1: Every admissible factor pair for $a = 2, 3, 4$.

The separate diagnostic range $a = 2, 3, 4$, $1 \leq b \leq 30$ contains 90 cubics, each with a noninteger root certified in an interval between consecutive integers. The reducible cases in that range are $b = 2, 6$ for $a = 2$, $b = 3, 12$ for $a = 3$, and $b = 4, 20$ for $a = 4$. Each has a linear and an irreducible

(a, b)	$f_{a,b}(x)$	Witness prime
$(2, 3)$	$x^3 - 47x^2 + 702x - 3312$	5
$(3, 10)$	$x^3 - 187x^2 + 11220x - 214200$	13
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(4, 21)$	$x^3 - 485x^2 + 75492x - 3721536$	5

Table 2: Root-free modular reductions for the exceptional cubics.

quadratic factor. These observations do not require whole-cubic irreducibility. Indeed, for every $a \geq 1$,

$$\begin{aligned} f_{a,a}(x) &= (x - 2a(a+1))(x^2 - (5a^2 + 2a)x + 6a^4 + 3a^3), \\ f_{a,a(a+1)}(x) &= (x - 2a(a+1)^2)(x^2 - (3a^3 + 4a^2 + 2a)x + 2a^6 + 6a^5 + 7a^4 + 3a^3). \end{aligned}$$

Consequently no prime can make every diagnostic cubic irreducible: the monic factors in the diagonal family retain positive degree modulo every prime.

For the second family the quadratic discriminant is $a^2(a^4 + 4a + 4)$. For $a \geq 3$ the radicand is between a^4 and $(a^2 + 1)^2$, since $2a^2 - 4a - 3 = 2(a-3)^2 + 8(a-3) + 3 > 0$. At $a = 2$ it is 28, between 5^2 and 6^2 . Thus the quadratic is irreducible for every $a \geq 2$.

Remark 7.4. At $a = 1, b = 2$ the cubic is $(x-3)(x-6)(x-8)$, although the full graph has irrational antisymmetric eigenvalues $(13 \pm \sqrt{33})/2$. A direct construction of its 10 vertices gives

$$\det(xI - L) = x(x-10)(x-9)^3(x-7)^2(x-4)(x^2 - 13x + 34).$$

Thus a noninteger cubic root is a sufficient condition for graph nonintegrality; the converse cannot be used without the other blocks.

8 Exact verification and reproducibility

The infinite-family conclusions follow from the written proofs above. The accompanying computations use integer or rational arithmetic to verify embeddings, identities and finite certificates. In particular, the all- b result for $a = 2, 3, 4$ uses the exhaustive divisor criterion, rather than extrapolation from the 90 diagnostic cases. No Lean formalization is claimed for these Laplacian results.

The code and saved JSON certificates are available in the public project repository. The build instructions beside this manuscript give commands and record the exact source revisions. The checks have the following scopes.

1. `scripts/check_integrity_obstructions.py` constructs ideals as exponent tuples and tests adjacency by coordinatewise maxima, independently of the support description. It verifies $LU = UB$ for the displayed invariant embeddings, for squarefree $t = 3, \dots, 8$ and $(1, 1, k)$ with $k = 1, \dots, 8$: 14 cases and 44 294 vertex pairs. It also checks the square-gap inequalities and principal-minor witnesses.
2. `scripts/check_remaining_case.py` constructs all 34 vertices for $(2, 2, 3)$ and verifies its full characteristic polynomial and the class decomposition in exact arithmetic. The shifted symmetric cubic is $x^3 - 80x^2 + 2099x - 18052$; its reduction modulo 5 has no root.

3. `scripts/check_repeated_exponent_family.py` checks the general quotient polynomial, both boundary sign identities, the positive-coefficient expansion, and the integral antisymmetric eigenvalues symbolically. Direct graphs for $(2, 2, 3)$, $(3, 3, 10)$, $(4, 4, 21)$ have 34, 174, 548 vertices, covering 165 490 vertex pairs. Exact neighbor counts verify both blocks and their lifting. The check computes the seven-support quotient polynomial; it does not compute the full 548×548 characteristic polynomial.
4. `scripts/check_aa_b.py` factors the 90 diagnostic cubics over $\mathbb{Q}$ and uses exact rational root isolation for a root between consecutive integers. Independent Sturm counts and endpoint signs were checked for every saved interval. Its modular search is restricted to 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31; absence of a listed witness is not a proof that no prime works for that cubic.
5. `scripts/check_fixed_repeated_exponents.py` reconstructs the cubic from B , checks the discriminant, product, cutoff and reducible factor identities, and enumerates every factor pair for $a = 2, 3, 4$. It verifies the admissibility conditions and all modular witnesses in Tables 1 and 2. The general congruence and factor inequality are proved in Section 6.
6. `scripts/check_second_cutoff.py` checks the polynomial remainder modulo $a + 2$, enumerates every divisor of 24, and verifies the four root-free cubic reductions settling all $d = a + 2$ cases. It also checks the second-cutoff identities and the two nonsquare discriminants completing $a = 5, 6$. This finite certificate follows the proved divisor reduction; it is not a parameter-range search.
7. `scripts/check_factor_index_reduction.py` verifies the general fixed-index divisibility and reconstruction identities, exhausts the divisors of $K_2 = 108$, and certifies the unique positive- b index-2 cubic modulo 13. It checks the third-cutoff identities and the four remaining discriminants for $a = 7, 8, 9$. These finite checks exhaust the proved candidate sets, with no arbitrary parameter cutoff.

The symbolic and characteristic-polynomial checks use SymPy 1.14.0; the first checker uses the Python standard library. Saved results contain source hashes. The 10-vertex polynomial in the preceding remark was independently checked by a direct ideal-graph construction; it is also reproducible by the command in the manuscript build guide. No floating-point eigenvalue calculation is used as integrality evidence.

9 Remaining questions

For repeated exponents $a \geq 10$, the next question is whether the symmetric cubic $f_{a,b}$ always has a noninteger root on the admissible square-discriminant pairs with

$$1 \leq b \leq \left\lfloor \frac{2(a-4)(a-2)}{3a+4} \right\rfloor.$$

Their smaller factor satisfies $d \geq 3a + 4$, with index $j \geq 3$. The two divisor criteria give finite problems for each fixed a or each fixed j , but the resulting family over arbitrary parameters is not settled. The cases with three pairwise distinct exponents, and nonsquarefree exponent vectors with more than three prime factors, require further arguments. The results here therefore do not establish nonintegrality for every integer with at least three distinct prime factors.

A useful direction is to combine the arithmetic conditions on (d, e) with a sign or congruence obstruction for the symmetric cubic. The distinction between the two blocks is essential: an integral antisymmetric block occurs on the boundary family, while an integral symmetric cubic occurs at $(1, 1, 2)$.

References

- [1] I. Chakrabarty, S. Ghosh, T. K. Mukherjee and M. K. Sen, *Intersection graphs of ideals of rings*, Discrete Mathematics **309** (2009), 5381–5392. doi:10.1016/j.disc.2008.11.034.